Ultra-Low-Power TinyML for Industrial Pump Cavitation Detection Under Extreme Acoustic Noise

Wei Zhao, Fatima Al-Hassan, Carlos Mendez

Published: International Conference on Industrial Cyber-Physical Systems (2025)

ABSTRACT

Centrifugal pump cavitation causes rapid impeller erosion and catastrophic system failure in chemical processing plants. We present CavitaNet, an ultra-low-power TinyML architecture running directly on a sub-dollar ARM Cortex-M4 microcontroller. CavitaNet operates within a 42 mW power budget and achieves 91.8% detection accuracy under standard workshop conditions. However, when ambient factory floor noise exceeds 78 dB, the precision collapses to 58.1% due to static spectral filtering limits, highlighting an urgent need for adaptive on-device noise cancellation.

1. Introduction

Continuous acoustic monitoring of slurry and cooling pumps provides instantaneous warning of cavitation inception. Deploying tethered power cables in hazardous chemical facilities is cost-prohibitive. Self-powered TinyML vibration sensors enable ubiquitous, wireless plant retrofitting.

2. TinyML Hardware & Model Pruning

We designed an ultra-compact Depthwise Separable CNN with only 24,000 parameters. Structured channel pruning and weight clustering reduced memory requirements to 48 KB RAM and 128 KB Flash.

3. Experimental Results

Our INT8 quantized convolutional neural network consumes only 42 mW on an ARM Cortex-M4 core, maintaining 91.8% cavitation detection accuracy under standard workshop conditions. The device successfully ran continuously for 9 months on a primary lithium coin cell.

4. Acoustic Noise Fragility

However, the classifier experiences severe false positive degradation (precision collapses to 58.1%) when ambient factory floor noise exceeds 78 dB, because the static spectral filter cannot adapt dynamically to fluctuating background interference (such as overhead cranes or adjacent diesel generators). Furthermore, zero on-device continual adaptation mechanism exists to handle mechanical aging drift.

5. Concluding Remarks

Sub-50mW TinyML is technically viable for pump diagnostics. Developing adaptive edge noise cancellation and continual few-shot learning remains imperative for deployment in noisy real-world industrial environments.